Chinese Journal of Chemistry, **2004**, 22, 990—993 **Article**

Effect of EtOH on the *E*-Iodohydroxylation of 1,2-Allenylic Sulfoxides with I_2[†]

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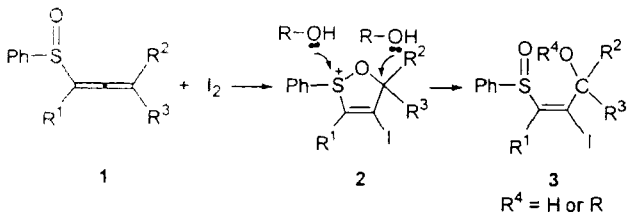
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The *E*-iodohydroxylation of 1,2-allenylic sulfoxides with I_2 in the presence of EtOH afforded 3-phenylsulfinyl-2-iodo-2(*E*)-alkenols in good yields and regio/stereo-selectivities under very mild conditions.

Keywords electrophilic addition, iodohydroxylation, allene, sulfoxide

Allenes are a class of versatile compounds for organic synthesis.[1,2] Recently we have observed the highly selective *E*-iodohydroxylation of 1,2-allenylic sulfoxides[3,4] affording *E*-3-phenylsulfinyl-2-iodo-2-alkenols in high yields.[5] The reaction of substituted 1,2-allenyl phenyl sulfoxides was usually carried out at r.t. while that of 1,2-propadienyl phenyl sulfoxide should be conducted at 55 ℃. In addition, it should be noted that the stereoselectivity of this reaction was determined by the formation of the cyclic intermediate **2**. Thus, we were interested to see the effect of EtOH on the structure of the final products (Scheme 1).

Scheme 1



When we ran this reaction with EtOH, it was quite interesting to observe that the reaction of 1,2-propadienyl phenyl sulfoxide (**1a**) with I_2 could proceed at 0 ℃. The *E*-iodohydroxylation product **4a** was formed in 58% yield with 27% of **1a** being recovered. Through careful analysis of the reaction mixture, it was observed that sulfide **5a** was also formed in 6%—10% yield (entries 1 and 2, Table 1). With 2 equiv. of I_2 and 20 equiv. of EtOH the reaction of **1a** with I_2 at 0 ℃ afforded **4a** in 87% yield (entry 7, Table 1). When the reaction was carried out at a higher temperature the yield of **5a** increased dramatically (entries 8—10, Table 1). Control experiment showed that the reaction in the absence of EtOH afforded *E*-**4a**[5] in 58% yield with 17% of **1a** being recovered (entry 12, Table 1). It should be noted that under the reaction conditions used in Ref. 5, *E*-**4a** was formed in 80% yield with 20% of **1a** being recovered (entry 13, Table 1). Under the current reaction conditions, the addition of LiOAc is not necessary.

This effect of EtOH on the rate of the *E*-iodohydroxylation reaction is quite general for different mono-, di-, and tri-substituted 1,2-allenylic sulfoxides[6] with the results summarized in Table 2. The reaction is very fast at 0 ℃ and the yields of **4** are from moderate to excellent while those of **5** are in the range of 6%—14%.

In order to study the effect of different alcohol on the iodohydroxylation of **1a**, benzyl alcohol, *i*-PrOH, or *n*-pentanol was used instead of EtOH. Although the yield of **4a** was comparable with **5a** being formed in 3%—9% yield, the reaction was not as clean as what we observed with EtOH (Table 3).

In conclusion, we have observed an interesting effect of alcohol on the rate of *E*-iodohydroxylation of 1,2-allenylic sulfoxides and I_2. Further studies in this area are being pursued in our laboratory.

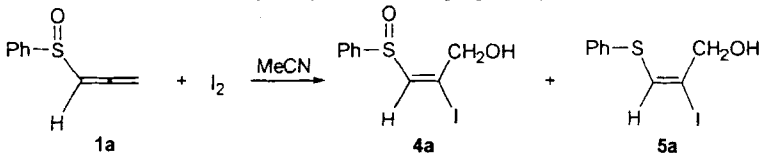
Experimental

Synthesis of *E*-2-iodo-3-phenylsulfinyl-2-propen-1-ol (4a)[5a]

To a solution of **1a** (82 mg, 0.5 mmol) in MeCN (2 mL) was added I_2 (254 mg, 1 mmol). After 5 min, a solution of EtOH (0.3 mL) in MeCN (4 mL) was added

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† Dedicated to Professor Chengye Yuan on the occasion of his 80th birthday.

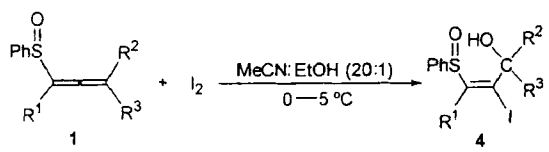
Table 1 Iodohydroxylation of 1,2-propadienyl sulfoxide[a]



Entry	I_2 (equiv.)	C_2H_5OH/mL	Temp./℃	Time/h	Yield[b]/%		Recovered/%
					4a	5a	1a
1	1.5	0.1	0—5	1.3	58	6	27
2	1.5	0.2	0—5	2.5	52	10	16
3	2	0.1	0—5	1.3	71	5	23
4	2	0.2	0—5	1.3	71	4	6
5	1.5	3	0—5	2	43	17	25
6	2	0.25	0—5	1.6	82	11	0
7	2	0.3	0—5	1.3	88	13	0
8	2	0.25	16	1.6	64	31	0
9	1.5	0.6	15	2	62	25	0
10[c]	2	0.25	16	1.6	45	46	0
11	2.4	0.3	55	1	11	0	0
12	1.5	0	0—5	2	58	14	17
13[d]	2	0	0—5	2	80	0	20

[a] The reaction was conducted using **1a** (0.5 mmol) in 6 mL of MeCN. [b] Isolated yield. [c] The reaction was carried out under an atmosphere of N_2. [d] The reaction was carried out with the addition of $LiOAc{\cdot}2H_2O$ (1 mmol) in $CH_3CN/H_2O=7:1$.

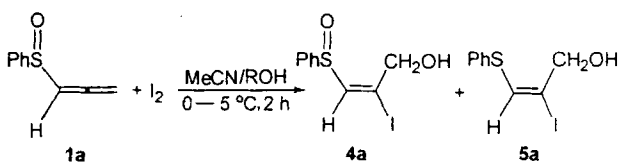
Table 2 Iodohydroxylation of 1,2-allenyl sulfoxides in MeCN-EtOH[a]



Entry	1			Time/h	Yield[b]/%
	R^1	R^2	R^3		4
1[c]	H	H	H (**1a**)	1.3	88 (**4a**)
2	H	H	*n*-C_4H_9 (**1b**)	0.7	84 (**4b**)
3	H	H	*n*-C_7H_{15} (**1c**)	0.5	82 (**4c**)
4	H	C_2H_5	C_2H_5 (**1d**)	0.3	91 (**4d**)
5	H	CH_3	*i*-C_4H_9 (**1e**)	0.3	80 (**4e**)
6	H	$(CH_2)_5$ (**1f**)		0.3	94 (**4f**)
7	H	CH_3	CH_2CH_3 (**1g**)	0.6	91 (**4g**)
8	H	CH_3	CH_3 (**1h**)	0.3	87 (**4h**)
9[d]	*n*-C_4H_9	CH_3	CH_3 (**1i**)	1	74 (**4i**)
10	*n*-C_4H_9	H	H (**1j**)	1	58 (**4j**)
11	H	CH_3	*n*-C_3H_7 (**1k**)	0.5	84 (**4k**)

[a] Condition A: To a solution of **1** (0.4 mmol) and EtOH (0.3 mL) in MeCN (2 mL) was added a solution of I_2 (0.8 mmol) in MeCN (4 mL). Before chromatography on silica gel, the reaction mixture was treated with $Na_2S_2O_3$ to remove the excess I_2. [b] Isolated yield. [c] To a solution of **1a** (0.5 mmol) in anhydrous MeCN (2 mL) was added I_2 (1 mmol). After 5 min, a solution of EtOH (0.3 mL) in MeCN (4 mL) was added with stirring. [d] The reaction mixture was submitted directly to chromatography on silica gel (eluent: petroleum ether/ethyl acetate=3 : 1, *V* : *V*) without the prior treatment of $Na_2S_2O_3$. With the treatment of $Na_2S_2O_3$, only the starting material **1i** was obtained.

Table 3 Effect of ROH on the iodohydroxylation of **1a**[a]



Entry	R	Yield[b]/% of **4a**
1	Bn	88
2	*i*-C_3H_7	85
3	*n*-C_5H_{11}	86

[a] The reaction was conducted using **1a** (0.5 mmol) and I_2 (1.2 mmol) in MeCN (6 mL)-ROH (0.3 mL). [b] Isolated yield.

with stirring. After being stirred at 0—5 ℃ for 1.3 h, the mixture was quenched with 4 mL of water followed by the addition of a saturated aqueous solution of $Na_2S_2O_3$. The mixture was extracted with dichloromethane (25 mL×3), washed with NaCl, and dried over $MgSO_4$. Evaporation and column chromatography on silica gel (eluent: petroleum ether/ethyl acetate=2 : 1, *V* : *V*) afforded **4a** (136.0 mg, 88%) as a white solid. m.p. 115—117 ℃ (dichloromethane-petroleum ether) (lit.[5a] 115.5—116.5 ℃); ^1H NMR ($CDCl_3$, 400 MHz) δ: 7.69—7.71 (m, 2H), 7.54—7.57 (m, 3H), 6.91 (s, 1H), 4.82 (t, J=6.4 Hz, 1H), 4.70 (dd, J=15.2, 6.4 Hz, 1H), 4.55 (dd, J=15.2, 6.4 Hz, 1H).

Synthesis of *E*-1-[1'-iodo-2'-(phenylsulfinyl)vinyl]-1-cyclohexanol (4f)[5a]

Typical procedure A To a solution of **1f** (69.6 mg, 0.3 mmol) and EtOH (0.3 mL) in MeCN (3 mL) was added a solution of I_2 (152.4 mg, 0.6 mmol) in

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MeCN (3 mL) with stirring at 0 ℃. After being stirred at 0 ℃ for 0.3 h, the mixture was quenched with 6 mL of water at this temperature. While the temperature rose to 0 ℃, a saturated aqueous solution of $Na_2S_2O_3$ was added. The mixture was extracted with dichloromethane (25 mL×3), washed with NaCl, and dried over $MgSO_4$. Evaporation and column chromatography on silica gel (eluent: petroleum ether/ethyl acetate=3 : 1, *V* : *V*) afforded **4f** (106.4 mg, 94%) as a white solid. m.p. 173—174 ℃ (dichloromethane-petroleum ether) (lit.[5a] 173—174 ℃); ^{1}H NMR ($CDCl_3$, 400 MHz) δ: 7.81—7.83 (m, 2H), 7.49—7.54 (m, 3H), 6.78 (s, 1H), 4.50 (s, 1H), 2.14—1.84 (m, 2H), 1.77—1.42 (m, 6H), 1.19—1.15 (m, 2H).

Following compounds were prepared according to typical procedure A.

***E*-2-Iodo-1-phenylsulfinyl-1-hepten-3-ol (*E*-4b)**[5b]: The reaction of 66 mg (0.3 mmol) of **1b** and 152.4 mg (0.6 mmol) of I_2 afforded 91.4 mg (84%) of **4b**. White solid, m.p. 107—108 ℃ (dichloromethane-petroleum ether) (lit.[5b] 107—108 ℃); ^{1}H NMR ($CDCl_3$, 400 MHz) δ: 7.62—7.69 (m, 2H), 7.50—7.55 (m, 3H), [7.00 (s, 0.72H), 6.93 (s, 0.28H)], 4.82 (bs, 1H), [4.47 (t, *J*=6.8 Hz, 0.28H), 4.41 (t, *J*=6.8 Hz, 0.72H), 1.66—1.74 (m, 1H), 1.33—1.50 (m, 5H), 0.90 (t, *J*=7 Hz, 3H).

***E*-2-Iodo-1-phenylsulfinyl-1-decen-3-ol (*E*-4c)**: The reaction of 131 mg (0.5 mmol) of **1c** and 254 mg (1 mmol) of I_2 afforded 167.1 mg (82%) of **4c**. White solid, m.p. 64—66 ℃ (dichloromethane-petroleum ether); ^{1}H NMR ($CDCl_3$, 400 MHz) δ: 7.60—7.68 (m, 2H), 7.49—7.53 (m, 3H), [6.98 (s, 0.58H), 6.91 (s, 0.42H)], 4.98 (bs, 1H), [4.45 (t, *J*=6.4 Hz, 0.42H), 4.39 (t, *J*=6 Hz, 0.58H)], 1.62—1.71 (m, 1H), 1.27—1.49 (m, 11H), 0.87 (t, *J*=6 Hz, 3H); IR (KBr) *v*: 3406, 1632, 1585, 1443, 1021 cm^{-1}; MS (70 eV, EI) *m/z* (%): 407 (M^++1, 4.95), 127 (100). Anal. calcd for $C_{16}H_{23}SO_2I$: C 47.30, H 5.71; found C 47.40, H, 5.86.

***E*-3-Ethyl-2-iodo-1-phenylsulfinyl-1-penten-3-ol (*E*-4d)**: The reaction of 88 mg (0.4 mmol) of **1d** and 203.2 mg (0.8 mmol) of I_2 afforded 132.4 mg (91%) of **4d**. White solid, m.p. 137—138 ℃ (dichloromethane-petroleum ether); ^{1}H NMR ($CDCl_3$, 400 MHz) δ: 7.75—7.78 (m, 2H), 7.49—7.52 (m, 3H), 6.96 (s, 1H), 2.98 (s, 1H), 1.93—2.08 (m, 2H), 1.61—1.73 (m, 2H), 1.01 (t, *J*=10 Hz, 3H), 0.76 (t, *J*=10 Hz, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 147.8, 145.5, 130.9, 129.2, 125.2, 118.2, 82.9, 34.33, 33.1; IR (KBr) *v*: 3421, 3198, 1581, 1461, 1442, 1015 cm^{-1}; MS (70 eV, EI) *m/z* (%): 365 (M^++1, 8.30), 335 (100). Anal. calcd for $C_{13}H_{17}O_2SI$: C 42.87, H 4.70; found C 42.87, H 4.85.

***E*-2-Iodo-3,5-dimethyl-1-phenylsulfinyl-1-hexen-3-ol (*E*-4e)**: The reaction of 93.6 mg (0.4 mmol) of **1e** and 203.2 mg (0.8 mmol) of I_2 afforded 121.5 mg (80%) of **4e**. White solid, m.p. 112—113 ℃ (dichloromethane-petroleum ether); ^{1}H NMR ($CDCl_3$, 400 MHz) δ: 7.77—7.80 (m, 2H), 7.48—7.50 (m, 3H), 6.80 (s, 1H), 4.50 (bs, 1H), 1.57—1.69 (m, 6H), 0.90 (d, *J*=6 Hz, 3H), 0.84 (d, *J*=6 Hz, 3H); IR (KBr) *v*: 3363, 2975, 1653, 1382, 1090, 1050 cm^{-1}; MS (70 eV, EI) *m/z* (%): 379 (M^++1, 13.62), 43 (100); HRMS (EI) calcd for $C_{14}H_{19}O_2SI$: 378.01505. found 378.01740.

***E*-2-Iodo-3-methyl-1-phenylsulfinyl-1-penten-3-ol (*E*-4g)**: The reaction of 81.7 mg (0.4 mmol) of **1g** and 203.2 mg, (0.8 mmol) of I_2 afforded 132.3 mg (91%) of **4g**. White solid, m.p. 120—121 ℃ (dichloromethane-petroleum ether); ^{1}H NMR ($CDCl_3$, 400 MHz) δ: 7.78—7.82 (m, 2H), 7.54—7.52 (m, 3H), 6.84 (s, 0.47 H), 6.83 (s, 0.53 H), 4.28 (s, 0.47H), 3.82 (s, 0.53H), 1.81—2.06 (m, 2H), 1.70 (s, 1.41H), 1.62 (s, 1.59H), 1.02 (t, *J*=7.6 Hz, 1.59H), 0.78 (t, *J*=7.6 Hz, 1.41H); IR (KBr) *v*: 3252, 1632, 1578, 1443, 1007 cm^{-1}; MS (70 eV, EI) *m/z* (%): 351 (M^++1, 8.20), 321 (100). Anal. calcd. for $C_{12}H_{15}O_2SI$: C 41.14, H 4.28; found: C 41.20, H 4.21.

***E*-3-Iodo-2-methyl-4-phenylsulfinyl-3-buten-2-ol (*E*-4h)**[5a]: The reaction of 76.8 mg (0.4 mmol) of **1h** and 203.2 mg (0.8 mmol) of I_2 afforded 117.2 mg (87%) of **4h**. White solid, m.p. 147—149 ℃ (dichloromethane-petroleum ether) (lit.[5a] 147.5—148.5 ℃); ^{1}H NMR ($CDCl_3$, 400 MHz) δ: 7.82—7.84 (m, 2H), 7.52—7.56 (m, 3H), 6.76 (s, 1H), 4.53 (bs, 1H), 1.69 (s, 3H), 1.47 (s, 3H).

***E*-3-Iodo-2-methyl-4-phenylsulfinyl-3-octen-2-ol (*E*-4i)**[5a]: The reaction of 74.4 mg (0.3 mmol) of **1i** and 114.3 mg, (0.45 mmol) of I_2 afforded 85.7 mg (74%) of **4i**. White solid, m.p. 150—152 ℃ (dichloromethane-petroleum ether). (lit[5a] 150—152 ℃); ^{1}H NMR ($CDCl_3$, 400 MHz) δ: 7.83—7.84 (m, 2H), 7.44—7.48 (m, 3H), 4.18 (s, 1H), 2.54—2.58 (m, 1H), 2.41—2.45 (m, 1H), 1.79 (s, 3H), 1.67 (s, 3H), 1.43—1.48 (m, 1H), 1.22—1.28 (m, 2H), 0.94—1.03 (m, 1H), 0.80 (t, *J*=6 Hz, 3H); MS *m/z* (%): 364 (M^++1, 47.48), 41 (100).

***E*-2-Iodo-3-phenylsulfinyl-2-hepten-1-ol (*E*-4j)**[5a]: The reaction of 66 mg (0.3 mmol) of **1j** and 152.4 mg, (0.6 mmol) of I_2 afforded 65.2 mg (58%) of **4j**. White solid, m.p. 83—84 ℃ (dichloromethane-petroleum ether) (lit[5a] 83—84 ℃); ^{1}H NMR ($CDCl_3$, 400 MHz) δ: 7.62 (d, *J*=4.8 Hz, 2H), 7.48—7.51 (m, 3H), 4.96 (d, *J*=10.8 Hz, 1H), 4.84 (d, *J*=10.8 Hz, 1H), 3.50 (bs, 1H), 2.41—2.48 (m, 1H), 2.13—2.19 (m, 1H), 1.40—1.44 (m, 1H), 1.16—1.26 (m, 2H), 0.65—0.85 (m, 4H).

***E*-2-Iodo-3-methyl-1-phenylsulfinyl-1-hexen-3-ol (*E*-4k)**: The reaction of 88.2 mg (0.4 mmol) of **1k** and 203.2 mg (0.8 mmol) of I_2 afforded 127.1 mg (84%) of **4k**. White solid, m.p. 123—124 ℃ (dichloromethane-petroleum ether); ^{1}H NMR ($CDCl_3$, 400 MHz) δ: 7.69—7.72 (m, 2H), 7.43—7.44 (m, 3H), 6.74 (s, 1H), 4.19 (s, 1H) 1.68—1.75 (m, 2H), 1.57 (s, 3H), 1.21—1.24 (m, 1H), 1.07—1.09 (m, 1H), 0.78 (t, *J*=6 Hz, 3H); IR (KBr) *v*: 3421, 1632, 1579, 1446, 1019 cm^{-1}; MS (70 eV, EI) *m/z* (%): 364 (M^+, 1.00), 321 (100). Anal. calcd. for $C_{13}H_{17}O_2SI$: C 42.87, H 4.70; found C 42.95, H 4.92.

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